6. Give the structure of CO and CO_2.
7. Give the uses of silicones.
8. Describe the structure of diborane.
9. Write a short note on hydroboration.
10. Give one example for each of the following

    (i) icosagens (ii) tetragens (iii) pnictogens (iv) chalcogens

11. Write a note on metallic nature of p-block elements.

12. Complete the following reactions

    a. $B(OH)_3 + NH_3 \longrightarrow$

    b. $Na_2B_4O_7 + H_2SO_4 + H_2O \longrightarrow$

    c. $B_2H_6 + 2NaOH + 2H_2O \longrightarrow$

    d. $B_2H_6 + CH_3OH \longrightarrow$

    e. $BF_3 + 9\ H_2O \longrightarrow$

    f. $HCOOH + H_2SO_4 \longrightarrow$

    g. $SiCl_4 + NH_3 \longrightarrow$

    h. $SiCl_4 + C_2H_5OH \longrightarrow$

    i. $B + NaOH \longrightarrow$

    j. $H_2B_4O_7 \xrightarrow{\text{Red hot}}$

13. How will you identify borate radical?

14. Write a note on zeolites.

15. How will you convert boric acid to boron nitride?

16. A hydride of 2nd period alkali metal (A) on reaction with compound of Boron (B) to give a reducing agent (C). identify A, B and C.

17. A double salt which contains fourth period alkali metal (A) on heating at 500K gives (B). Aqueous solution of (B) gives white precipitate with $BaCl_2$ and gives a red colour compound with alizarin. Identify A and B.

18. CO is a reducing agent. Justify with an example.

XII_U2-P-Block.indd 55 22-12-2022 20:06:04